

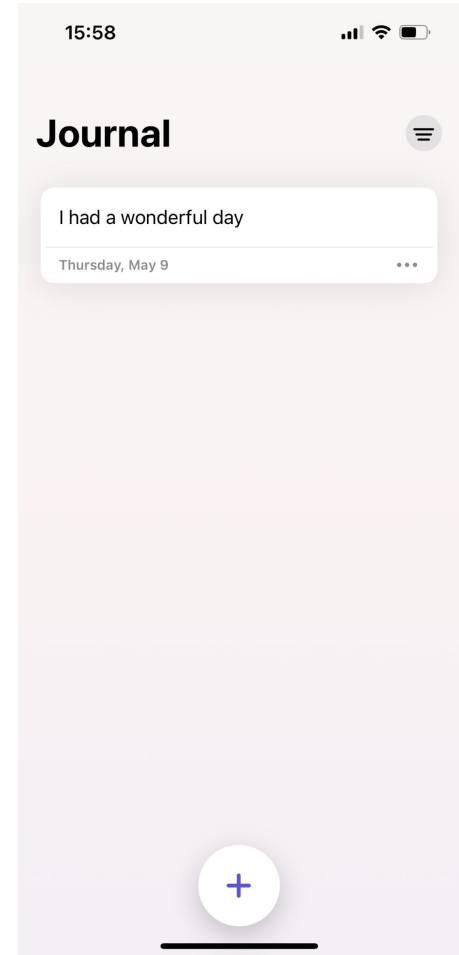
Personal Blog

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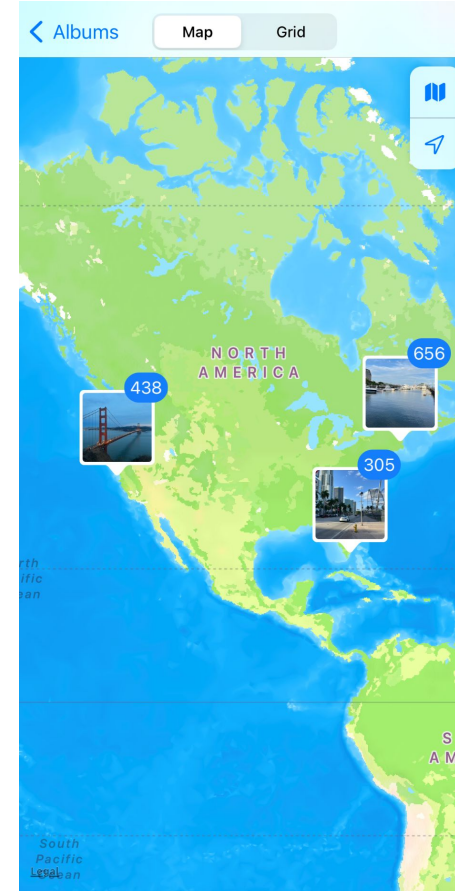
Motivation

- A personal blog app with more interesting features inspired from other apps



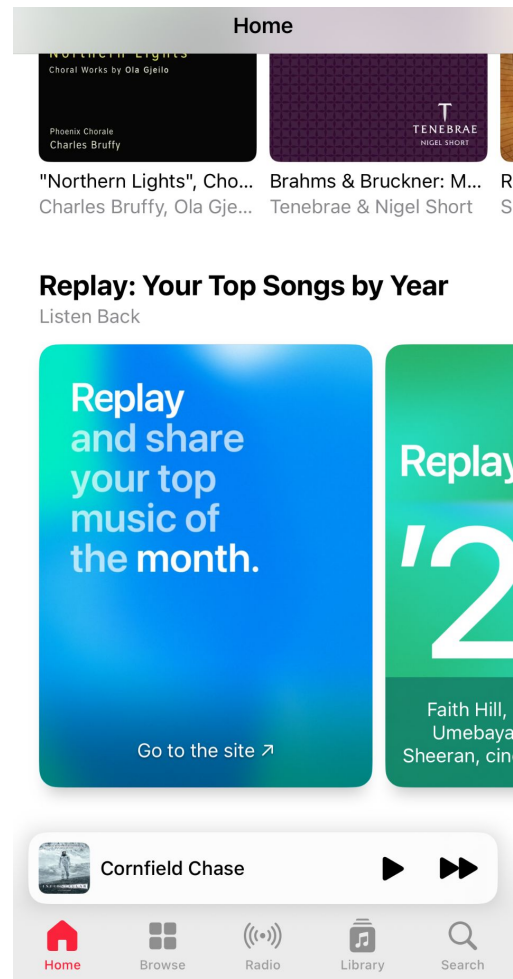
Motivation: new feature for blog

- Interactive map showing the locations recorded in journals
- Inspired by map feature of photo app



Motivation: Memories

- Collection of historical blogs of specific time range to refresh your precious memories
- Inspired by the replay feature from Apple Music



Demo

Roadblocks

Creating MapPage through activity and integrating it with Compose

Key Problems:

- Navigating between Pages
- Compose will not recompose when update was made through Activity unless you specifically tell it to do so
- Facing difficulties when calling activity function and Compose functions because special restrictions

Workarounds: Fragment v.s. composable

- To create the map screen and select locations from it, we tried to create a Map Activity with a `activity_map.xml` first, while we've done all the other part with Jetpack Composables
- The challenge that drove us back to composable: when returning from MapActivity, all the composable functions are not updated immediately. They cannot get information from intent as easily as another activity.

Workarounds: Map Activity

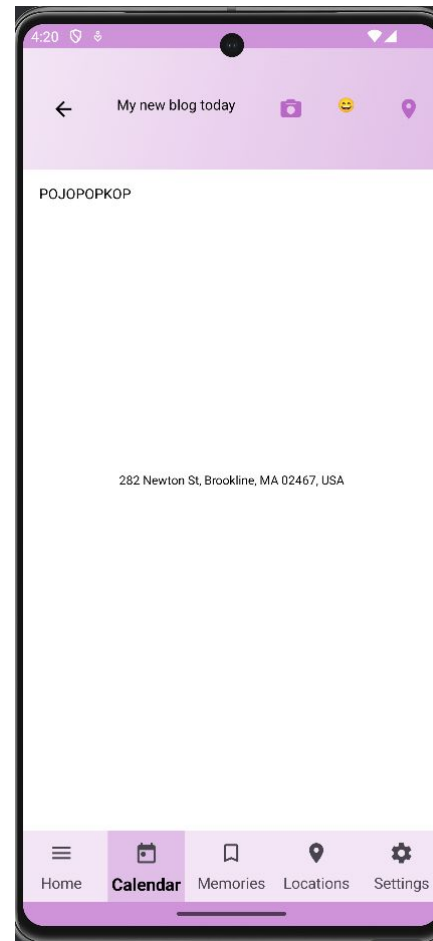
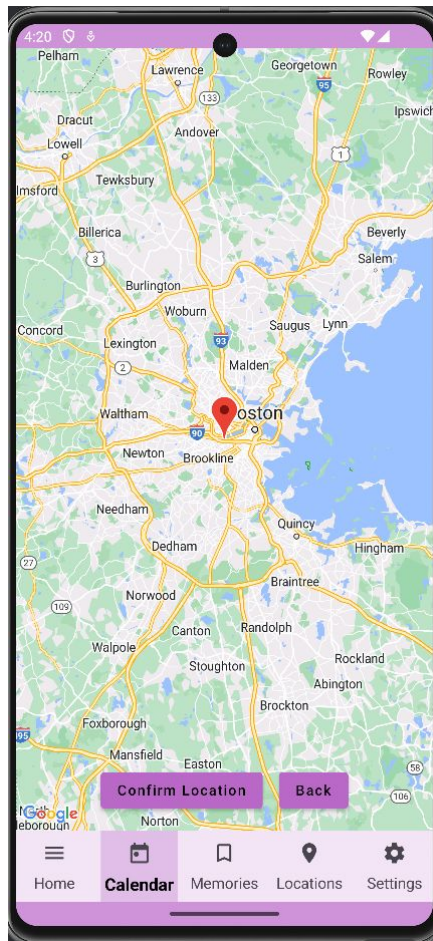
- The method to get location in a Map Activity and update the database
- Problem: the composables will not update immediately.

```
val blogDetailViewModel: BlogDetailViewModel by viewModels {
    BlogDetailViewModelFactory(uuid)
}

lifecycleScope.launch { this: CoroutineScope
    repeatOnLifecycle(Lifecycle.State.STARTED) { this: CoroutineScope
        blogDetailViewModel.blog.collect { blog ->
            if (blog != null) {
                Log.d( tag: "update address", msg: "${blog.id}")
            }
            blogDetailViewModel.updateBlog { blog ->
                Log.d( tag: "update address", address)
                blog.copy(location = address) ^updateBlog
            }
        }
    }
}

val returnIntent = Intent()

returnIntent.putExtra( name: "location_name", address)
returnIntent.putExtra( name: "blog_id", blogId)
setResult(Activity.RESULT_OK, returnIntent)
finish()
```

Workarounds: Map by composables

- NavController updates the page

```
onClick = {  
    location.let { it: State<LatLng>  
        locationviewModel.confirmLocation( description: "Location confirmed with description")  
        locationviewModel.setLocation(it.value)  
        val newAddress =  
            addresses?.firstOrNull()?.getAddressLine( index: 0) ?: "Unknown location"  
  
        blogDetailViewModel.updateBlog { blog ->  
            Log.d( tag: "update address", address)  
            blog.copy(location = newAddress) ^updateBlog  
        }  
        navController.popBackStack()  
    }  
},  
) { this: RowScope  
    Text( text: "Confirm Location")
```

What we could do differently

- Hold the calendar state
 - Each time we turn to calendar page it shows the page of today, but not the page of month you were looking at last time
- When got prompt into the map, we could try to use the phone's current Location and adjust map based on the current location.

Future work

- Swipe between screens / swipe to delete
- Interactive Map
 - Select blogs by clicking on the markers on the map
 - Markers showing address information and pictures (like Apple photos)
- Multiple photos /locations in one blogs
- Bind with personal account (e.g. App account & Instagram)
- Cloud Storage
- UI polish

References

<https://developers.google.com/maps/documentation/android-sdk/overview?hl=zh-cn>
<https://medium.com/@khater/swipeable-component-with-jetpack-compose-swipe-to-delete-or-archive-debfe4503613>
<https://developer.android.com/develop/ui/views/notifications/notification-permission>
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<https://github.com/kizitonwose/Calendar>
<https://chatgpt.com/>
<https://developer.android.com>

Thank you!